NAMMACONNECT

Product Requirements Document (PRD)

Product: NammaConnect — Context-Aware Kannada Learning and Communication Platform
Version: 1.0
Document Type: Product Requirements Document
Project Category: Regional Language Technology, Education, Accessibility
Primary Problem Statement: Kannada in Everyday Life
Secondary Alignment: Kannada for Everyone
Target Platform: Responsive Web Application
Initial Target Market: Karnataka, India
Project Stage: MVP Development

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1. Executive Summary

NammaConnect is a practical Kannada learning and communication platform designed to help people who do not speak Kannada fluently communicate more confidently in everyday situations.

The platform focuses on teaching Kannada through realistic scenarios rather than relying exclusively on conventional vocabulary lists and grammar lessons.

Users can select a situation, learn relevant Kannada phrases, understand their meanings, practise conversations, interpret common responses, and complete interactive exercises.

The initial version will focus on transportation, shopping, education, and basic healthcare-related communication.

The system will use a verified phrase database and predefined conversational workflows to maintain linguistic reliability. Language-processing and speech technologies may be integrated where they improve the learning experience.

The objective is to create a usable application that addresses a practical communication problem, demonstrates measurable learning outcomes, and can be expanded into a broader regional-language learning product.

2. Problem Statement

People who do not speak Kannada fluently may experience communication difficulties when travelling, shopping, attending educational institutions, visiting healthcare facilities, or interacting with local businesses in Karnataka.

Conventional language-learning methods may teach vocabulary and grammar without adequately preparing learners to handle practical conversations. Translation tools can help convert individual sentences, but they do not necessarily provide structured situational practice, explain conversational context, or help users understand likely responses.

Consequently, learners may know individual Kannada words but still struggle to use them appropriately in everyday interactions.

There is a need for an accessible, situation-based learning platform that helps non-Kannada speakers understand, practise, and use Kannada in real-world contexts.

3. Proposed Solution

NammaConnect will provide an interactive learning environment organized around everyday communication tasks.

Instead of starting with grammar lessons, users will begin by selecting a practical situation and identifying what they want to communicate.

The platform will then provide appropriate Kannada phrases, English meanings, transliteration, pronunciation assistance, contextual explanations, and guided conversation exercises.

Users will practise likely conversational exchanges and complete assessments to evaluate their understanding.

The application will also support saving frequently used phrases for quick reference.

The first version will prioritize verified language content, a simple user experience, and reliable functionality over unrestricted AI-generated conversations.

4. Product Vision

To make practical Kannada communication more accessible to people living, studying, working, or travelling in Karnataka.

5. Product Objectives

The product will aim to:

1. Help non-Kannada speakers learn useful expressions for everyday situations.
2. Reduce the difficulty of finding appropriate Kannada phrases during common interactions.
3. Improve learners' understanding of frequently encountered conversational responses.
4. Provide structured practice through realistic scenarios.
5. Make essential Kannada phrases available for quick reference.
6. Evaluate whether users improve their practical communication skills.
7. Establish a foundation for future speech-enabled regional-language assistance.

6. Target Users

6.1 Primary Users

A. Students from Other States

Students studying in Karnataka who are unfamiliar with Kannada and need to communicate with classmates, shopkeepers, transport providers, and college staff.

B. New Residents

People who have recently moved to Karnataka for employment, education, or personal reasons.

C. Visitors and Travellers

People who visit Karnataka and want to communicate in common situations without depending entirely on another person.

6.2 Secondary Users

- Employees joining organizations in Karnataka.
- People interested in learning practical Kannada.
- Educational institutions supporting students from different linguistic backgrounds.
- Organizations that want to help newcomers learn essential Kannada expressions.

7. User Personas

Persona 1: The New Student

Background: A student from Maharashtra joins a college in Dharwad.

Problem: The student struggles to communicate with local shopkeepers and auto drivers.

Goal: Learn essential Kannada phrases and understand common responses.

Expected benefit: The ability to practise conversations before using them in real situations.

Persona 2: The New Employee

Background: An employee relocates to Bengaluru for work.

Problem: Everyday interactions outside the workplace are difficult because the employee is unfamiliar with Kannada.

Goal: Learn useful expressions for transportation, shopping, and basic local communication.

Expected benefit: A structured way to learn relevant phrases without completing an extensive language course first.

Persona 3: The Short-Term Visitor

Background: A traveller visits Karnataka for several days.

Problem: The traveller needs essential phrases but does not have time to study the language extensively.

Goal: Quickly access and practise commonly used expressions.

Expected benefit: A searchable, mobile-friendly phrasebook with contextual explanations.

8. Product Scope

8.1 MVP Features

The first version will include:

1. User onboarding.
2. Situation-based learning categories.
3. Searchable Kannada phrases.
4. Kannada script, English meaning, and English transliteration.
5. Contextual explanations.

6. Guided conversation practice.
7. Interactive quizzes.
8. Progress tracking.
9. Saved phrases.
10. Offline access to a selected collection of essential phrases.

8.2 Features Excluded from MVP

The following will not be mandatory for the initial release:

- Unrestricted, open-ended AI conversations.
- Automatic pronunciation scoring.
- Continuous background listening.
- Real-time translation of arbitrary conversations.
- A complete Kannada grammar course.
- Advanced regional dialect classification.
- User-generated public phrase submissions.
- Social networking and community chat.
- Paid subscriptions and payment processing.

These exclusions keep the first version achievable and reduce dependence on complex speech and language models.

9. Functional Requirements

FR-01: User Onboarding

The system shall allow users to begin using the application through a simple onboarding flow.

The onboarding process should ask about the user's preferred explanation language and approximate Kannada proficiency.

Supported proficiency levels:

- Beginner.
- Basic conversational.
- Intermediate.

Users should be able to explore essential phrases without creating an account.

Authentication may be introduced for users who want synchronized progress and saved content across devices.

Acceptance criteria:

- Users can enter the application without a lengthy registration process.

- Users can select their explanation language.
- Users can change their preferences later.

FR-02: Situation-Based Categories

The application shall organize learning content into practical categories.

Initial categories:

1. Transportation.
2. Shopping and markets.
3. Education and college.
4. Basic healthcare communication.

Each category shall contain specific situations and communication goals.

Example:

Transportation → Auto-rickshaw → Tell the driver your destination.

Acceptance criteria:

- Users can browse the available categories.
- Selecting a category displays its supported scenarios.
- Each scenario contains relevant learning material.
- Empty categories are handled gracefully.

FR-03: Scenario Selection

The system shall allow users to select a communication task.

Example tasks:

- Ask where the bus stand is.
- Ask the price of an item.
- Request directions inside a college.
- Ask where a hospital department is located.

The scenario page shall explain the communication objective before presenting the phrases.

Acceptance criteria:

- Every published scenario has a clear objective.
- Users can move from a scenario to its learning content.
- Users can return to the category without losing their place.

FR-04: Kannada Phrase Display

The system shall display each phrase using a structured content format.

Each phrase should contain:

- Kannada text.
- English translation.
- English transliteration.
- Context or intended use.
- Formality or politeness information where relevant.
- Optional audio pronunciation.
- Related phrases where appropriate.

Example:

Situation: Asking for directions.

Kannada: ಬಸ್ ನಿಲ್ದಾಣ ಎಲ್ಲಿದೆ?

English meaning: Where is the bus stand?

Transliteration: Bus nildana ellide?

Context: A basic question used to ask someone for the location of a bus stand.

All published Kannada phrases must be reviewed for correctness and naturalness.

Acceptance criteria:

- Every published phrase has a Kannada sentence and English meaning.
- Transliteration is available for the supported phrases.
- The interface renders Kannada characters correctly.
- Phrase content can be corrected without changing the application code.

FR-05: Phrase Search

The system shall provide a search feature for finding useful phrases.

Users should be able to search by:

- English keywords.
- Communication goals.
- Situation names.
- Supported Kannada text.

Example search:

User searches for "price".

The application returns relevant phrases for asking about prices, quantities, and discounts.

Acceptance criteria:

- Search results display relevant phrases.
- No-result searches show a useful message.
- Users can open a result directly.
- Search functionality works with the supported language inputs.

FR-06: Contextual Explanation

The system shall explain when and why a phrase is used.

Explanations may include:

- Meaning of important words.
- Appropriate situations.
- Polite alternatives.
- Common variations.
- Related questions and answers.

The explanations should be written in accessible language suitable for beginners.

Acceptance criteria:

- Explanations are available for published learning phrases.
- The system distinguishes literal meaning from contextual usage when necessary.
- Explanations do not introduce unsupported cultural or linguistic claims.

FR-07: Guided Conversation Practice

The application shall provide predefined conversation flows for selected scenarios.

Example:

Scenario: Buying an item at a shop.

Step 1: The application presents a shopkeeper's question.

Step 2: The user selects or enters an appropriate response.

Step 3: The application explains why the response is suitable.

Step 4: The conversation continues to the next step.

Step 5: The user receives a summary of the interaction.

Conversation flows should include realistic alternative responses where appropriate.

The initial version may use selectable responses rather than speech recognition.

Acceptance criteria:

- Each supported conversation has a defined starting point.
- Users can complete the conversation using the available responses.

- Feedback explains incorrect answers.
- Users can restart or repeat the scenario.
- The system does not depend on unrestricted AI generation to complete predefined lessons.

FR-08: Interactive Practice and Quizzes

The system shall provide short assessments related to completed learning scenarios.

Supported question types:

- Select the correct Kannada phrase.
- Match Kannada phrases with English meanings.
- Select an appropriate response to a question.
- Identify the meaning of a common expression.

Each assessment shall calculate a score based on the answers provided.

Acceptance criteria:

- Questions have defined correct answers.
- Users receive feedback after submitting their responses.
- Scores are calculated consistently.
- Users can repeat an assessment.

FR-09: Progress Tracking

The system shall track learning activity for users who choose to save their progress.

Possible metrics:

- Scenarios completed.
- Quiz scores.
- Categories explored.
- Saved phrases.
- Recently practised scenarios.

Acceptance criteria:

- Progress reflects completed activities.
- Saved progress remains available after the user returns.
- The system does not falsely mark incomplete scenarios as completed.

Progress synchronization will require an account or an equivalent persistent storage mechanism.

FR-10: Saved Phrases

Users shall be able to bookmark phrases for quick access.

The saved collection should display the Kannada phrase, its meaning, and the scenario in which it is used.

Acceptance criteria:

- Users can save and remove phrases.
- Duplicate bookmarks are prevented.
- Saved phrases are available from a dedicated section.
- Offline availability is supported for downloaded or locally cached content.

FR-11: Offline Phrasebook

The system shall provide a predefined collection of essential phrases that can be accessed without an active internet connection after the content has been downloaded or cached.

The offline collection should prioritize common transportation, shopping, and emergency-assistance expressions.

Acceptance criteria:

- Downloaded phrases remain accessible offline.
- The application clearly indicates which content is available offline.
- Offline access does not depend on live AI APIs.
- New content can be synchronized when connectivity returns.

Offline support is limited to the content explicitly made available on the device; the entire application is not required to function offline.

FR-12: Content Management

The system shall support maintaining the phrase database and conversation scenarios.

Authorized maintainers should be able to add, edit, review, publish, or remove content.

Each published phrase should have a review status.

Suggested statuses:

- Draft.
- Under review.
- Approved.
- Published.
- Archived.

Acceptance criteria:

- Unapproved content cannot be published accidentally.
- Changes to phrases can be made without modifying frontend components.

- Only authorized maintainers can publish content.

10. Non-Functional Requirements

10.1 Usability

The interface must be understandable to users with limited Kannada knowledge.

Important information should be visible without requiring users to navigate multiple screens.

10.2 Performance

The application should aim to:

- Load common learning screens within approximately two seconds under reasonable network conditions.
- Return ordinary phrase-search results within one second for the initial dataset.
- Provide responsive interactions on typical mobile devices.

These are development targets and must be validated through testing.

10.3 Accessibility

The interface should:

- Use readable text sizes.
- Maintain sufficient contrast.
- Support keyboard navigation.
- Label interactive controls clearly.
- Render Kannada Unicode text correctly.
- Avoid relying on color alone to communicate feedback.

10.4 Security

The application must protect user data and restrict content-management access.

If accounts are introduced, authentication credentials must be handled securely.

10.5 Reliability

The learning experience should not fail because an optional AI or speech service is unavailable.

Core phrase browsing, predefined lessons, quizzes, and saved offline content should remain independent of external AI services wherever possible.

10.6 Maintainability

The application should use modular components and a structured content model so that additional scenarios and phrases can be introduced without redesigning the entire system.

11. User Workflow

Primary Learning Workflow

1. User opens NammaConnect.
2. User selects a preferred explanation language.
3. User selects a practical situation.
4. User chooses a communication task.
5. The application displays relevant Kannada phrases.
6. User reviews the meaning and pronunciation guide.
7. User starts a guided conversation.
8. User selects or enters responses.
9. The application provides feedback.
10. User completes a short assessment.
11. The application records the result if progress tracking is enabled.
12. User can save phrases or practise another scenario.

Quick Phrase Workflow

1. User opens the application.
2. User searches for a phrase or communication goal.
3. The application displays matching phrases.
4. User opens the relevant phrase.
5. User reviews the Kannada sentence and meaning.
6. User saves the phrase if needed.

Offline Workflow

1. User downloads the essential phrase collection while connected.
2. The application stores the supported content locally.
3. User opens the saved collection without internet connectivity.
4. User browses the available phrases.
5. The application synchronizes supported changes when connectivity returns.

12. System Architecture

The MVP will use a conventional web application architecture.

Frontend

Recommended: React with TypeScript, or Next.js with TypeScript.

Responsibilities:

- User interface.

- Category and scenario navigation.
- Phrase display and search.
- Conversation practice.
- Quiz interactions.
- Progress visualization.
- Offline content access.

Backend

Recommended: Python with FastAPI.

Responsibilities:

- Phrase and scenario APIs.
- Search requests.
- Learning-content retrieval.
- Progress management.
- Quiz scoring where server-side validation is needed.
- Authentication if introduced.
- Content-management operations.

Database

Recommended: PostgreSQL for a more extensible deployment, or SQLite for an initial local prototype.

Primary entities:

- Users.
- Categories.
- Scenarios.
- Phrases.
- Conversation flows.
- Conversation steps.
- Quiz questions.
- Quiz options.
- User progress.
- Saved phrases.
- Content reviews.

Language Technology

The first version should rely on verified phrase content and predefined conversation flows.

Optional technologies include:

- Text-to-speech for pronunciation.
- Speech recognition for spoken input.
- Translation services for additional language support.
- Language models for generating draft explanations or conversation variants.

Any AI-generated language content must be validated before being treated as authoritative learning material.

Offline Storage

The web application may use browser storage and service-worker caching to support offline access to selected content.

Deployment

The frontend and backend may be deployed separately or together, depending on the chosen hosting environment.

The system must be usable from a mobile browser without requiring users to install a native mobile application.

13. Suggested Data Model

Category

- category_id
- category_name
- description
- icon
- display_order
- status

Scenario

- scenario_id
- category_id
- title
- objective
- difficulty_level
- estimated_duration

- status

Phrase

- phrase_id
- scenario_id
- kannada_text
- english_meaning
- transliteration
- contextual_explanation
- formality_level
- pronunciation_audio_reference
- review_status
- reviewer_reference

Conversation

- conversation_id
- scenario_id
- title
- introduction
- difficulty_level
- status

ConversationStep

- step_id
- conversation_id
- step_order
- speaker_role
- kannada_text
- english_meaning
- expected_response_reference
- explanation

QuizQuestion

- question_id
- scenario_id

- question_text
- question_type
- explanation
- status

QuizOption

- option_id
- question_id
- option_text
- is_correct

UserProgress

- progress_id
- user_id
- scenario_id
- completion_status
- latest_score
- last_practised_at

SavedPhrase

- user_id
- phrase_id
- saved_at

The database should enforce relationships between categories, scenarios, phrases, conversations, and assessments.

14. AI and Machine Learning Strategy

AI should be introduced only where it improves a clearly identified function.

MVP Approach

The first version does not require a custom-trained machine-learning model.

Use:

- A verified phrase database.
- Rule-based scenario selection.
- Predefined conversation flows.
- Deterministic quiz scoring.

- Conventional search.
- Optional existing speech or language services after testing.

Future AI Capabilities

Potential extensions include:

1. Natural-language intent recognition to identify what the user wants to communicate.
2. Context-aware phrase recommendations.
3. Speech recognition for Kannada conversation practice.
4. Adaptive learning based on quiz performance.
5. Personalized revision suggestions.

AI Reliability Requirements

- Generated Kannada phrases must not be presented as verified without appropriate review.
- Speech-recognition results must be treated as potentially incorrect.
- The system must not claim that pronunciation is correct solely because speech recognition produced a transcript.
- Users should be able to access essential phrases without relying on generative AI.

15. Content Strategy

Content quality is central to the product's success.

The project should begin with a small, carefully reviewed dataset rather than a large collection of unverified phrases.

Suggested initial content target:

- Four learning categories.
- Five to ten scenarios per category.
- Ten to fifteen useful phrases per scenario.
- At least one guided conversation for each completed scenario.
- Short assessments for the supported scenarios.

These figures are planning targets, not existing dataset counts.

Content should be reviewed by fluent Kannada speakers familiar with natural everyday usage.

Where regional differences exist, the content should identify relevant variations rather than presenting one expression as universally preferred.

16. User Interface Requirements

The interface should be mobile-first, simple, and suitable for users who are unfamiliar with Kannada.

Main Screens

Screen 1: Home

Displays:

- Welcome message.
- Search field.
- Learning categories.
- Continue-learning option.
- Saved phrases.
- Recent activity.

Screen 2: Categories

Displays cards for transportation, shopping, education, and healthcare communication.

Screen 3: Scenario List

Displays practical tasks within the selected category.

Screen 4: Phrase Learning

Displays:

- Kannada phrase.
- English meaning.
- Transliteration.
- Contextual explanation.
- Audio playback where supported.
- Save button.
- Related phrases.

Screen 5: Conversation Practice

Displays the current conversational turn, possible responses, feedback, and navigation controls.

Screen 6: Quiz

Displays questions, response options, submission controls, and results.

Screen 7: Progress

Displays completed scenarios, recent quiz scores, and areas that need revision.

Screen 8: Saved Phrases

Displays bookmarked phrases and their categories.

Screen 9: Offline Phrasebook

Displays essential phrases available locally on the device.

17. Success Metrics

The project should be evaluated through actual user testing.

Product Metrics

- Number of users who complete an initial scenario.
- Scenario completion rate.
- Average time required to locate a phrase.
- Frequency of repeat practice.
- Quiz completion rate.
- Number of phrases saved.

Learning Metrics

- Pre-test and post-test score differences.
- Accuracy in selecting appropriate responses.
- Understanding of common conversational replies.
- Ability to recall learned phrases after a delay.

Language Quality Metrics

- Percentage of reviewed phrases accepted by Kannada-speaking reviewers.
- Number of identified translation or transliteration errors.
- User-reported confusion about phrase meaning or context.

Proposed Evaluation Method

Recruit a small group of participants who are unfamiliar with Kannada.

1. Give participants a short assessment of practical Kannada comprehension.
2. Ask them to complete selected learning scenarios using NammaConnect.
3. Repeat an equivalent assessment.
4. Compare the results.
5. Collect usability feedback.
6. Ask fluent Kannada speakers to review the phrases and conversations used in the evaluation.

The project should report the observed results rather than claiming improvements before testing.

18. Development Roadmap

Phase 1: Research and Validation

Tasks:

- Identify the primary user group.

- Interview potential users.
- Select the first practical scenarios.
- Collect candidate Kannada phrases.
- Arrange linguistic review.
- Define the evaluation method.

Deliverable: Validated requirements and an initial content specification.

Phase 2: Core Application

Tasks:

- Create the frontend structure.
- Implement category and scenario navigation.
- Build the phrase database.
- Implement phrase search.
- Display Kannada text, translations, and transliteration.

Deliverable: A functional phrase-learning application.

Phase 3: Learning and Practice

Tasks:

- Implement guided conversations.
- Create quizzes.
- Add feedback.
- Implement saved phrases.
- Add progress tracking.

Deliverable: A complete text-based learning workflow.

Phase 4: Offline Access and Testing

Tasks:

- Cache the selected phrase collection.
- Test offline browsing.
- Evaluate mobile responsiveness.
- Test language rendering.
- Correct identified defects.

Deliverable: A usable MVP suitable for demonstrations.

Phase 5: Evaluation

Tasks:

- Conduct user testing.
- Measure learning outcomes.
- Review phrase quality.
- Document limitations.
- Refine the user experience.

Deliverable: Evaluation results for the project report and demonstration.

Phase 6: Optional Enhancements

Tasks:

- Integrate speech playback.
- Test Kannada speech recognition.
- Explore pronunciation feedback.
- Add further scenarios based on user feedback.

Deliverable: An extended version supported by testing.

19. Risks and Mitigation

Risk 1: Incorrect Kannada Content

Problem: Incorrect phrases could confuse learners.

Mitigation: Require review before publishing learning content.

Risk 2: Scope Expansion

Problem: Attempting to support every real-world situation could delay the MVP.

Mitigation: Limit the initial release to four categories and a controlled set of scenarios.

Risk 3: Dependence on AI Services

Problem: External services may be unavailable, expensive, or inaccurate.

Mitigation: Keep essential functionality based on verified content and deterministic workflows.

Risk 4: Weak Differentiation

Problem: The product could resemble a conventional phrasebook.

Mitigation: Emphasize guided situational conversations, response comprehension, and measurable task completion.

Risk 5: Unreliable Pronunciation Assessment

Problem: Speech recognition does not necessarily provide reliable pronunciation scoring.

Mitigation: Treat pronunciation evaluation as an optional feature requiring independent validation.

Risk 6: Limited User Adoption

Problem: Users may not continue practising after their immediate need is addressed.

Mitigation: Offer quick phrase lookup, short lessons, bookmarks, and relevant scenario recommendations.

20. Privacy and Data Protection

The application should collect only the information necessary to deliver its features.

Users should be able to access basic phrase-learning functionality without creating an account.

If progress tracking is enabled, the application should explain what information is stored and how it is used.

If speech features are introduced, the product must clearly disclose whether audio is processed locally or transmitted to an external service.

Audio should not be retained by default unless retention is necessary, disclosed, and appropriately controlled.

21. Future Product Opportunities

Potential future extensions include:

- Kannada conversation practice using speech.
- More regional expressions and dialect variations.
- Additional explanation languages.
- Educational institution deployments.
- Workplace onboarding programs.
- Specialized learning paths for visitors and new residents.
- Carefully validated context-aware phrase suggestions.
- Offline learning packages for different locations and situations.

These extensions should be prioritized according to demonstrated user needs.

22. MVP Definition of Done

The MVP will be considered ready for evaluation when:

1. Users can browse the initial learning categories.
2. Users can open practical communication scenarios.
3. Each published phrase has reviewed Kannada text and an English meaning.
4. Users can view transliteration and contextual explanations.
5. Users can complete at least one guided conversation in every supported scenario.
6. Users can complete quizzes and receive consistent feedback.

7. Saved phrases work as specified.
8. The essential phrase collection is available offline after caching or downloading.
9. The application works on common mobile and desktop browsers.
10. The team has tested the application with representative users.
11. Language quality and learning outcomes have been evaluated.
12. Known limitations are documented.

23. Final Product Definition

NammaConnect is a situation-based Kannada learning and communication platform that helps non-Kannada speakers practise practical conversations for everyday life in Karnataka.

Its primary value is the connection between learning a phrase and understanding how to use it in a realistic interaction.

The first version will prioritize verified language content, guided conversations, comprehension exercises, and quick phrase access. Optional speech and AI capabilities will be introduced only when they can be implemented and evaluated reliably.

The project will be assessed on its ability to improve practical communication, the quality of its Kannada content, the usability of its workflows, and the evidence collected through user testing.

Primary alignment: Kannada in Everyday Life.

Secondary alignment: Kannada for Everyone.

Initial product principle: Help users communicate in practical situations before expanding into advanced language technology.